# FIKS: Turning Silent Tool Failures into Gated Repair Pull Requests for MCP Servers

System Design and Pilot Evaluation

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## Author review checklist

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**Abstract**

LLM agents' tools can fail silently: a Model Context Protocol (MCP) server returns a well-formed success whose content is wrong, for example after its upstream API changes behaviour but not its schema. Such failures occur and agents often adopt the wrong values; a schema checks shape, not meaning. Runtime monitors warn the current agent, but the tool stays broken for later ones. We present FIKS, which turns a silent tool failure into a reviewable fix. A proxy checks each successful MCP result against an outcome contract (including agreement with the raw upstream exchange and read-after-write) and returns a receipt to the agent. A violation becomes a deterministic reproduction test that must fail on the current code for the named property; a tool-restricted coding agent repairs the server; and a pull request is opened only after that test, the contract suite, stored traces, typecheck, lint and CI on the exact commit pass. FIKS never merges. In a pilot with 5 seeded drifts on a mock vendor whose OpenAPI document is unchanged, FIKS flagged all 5, built a valid fail-first test for each and got a green-CI pull request in one repair attempt per drift (1.4–1.6 min from violation to PR; \$0.11–\$0.21 per repair at API prices). In a 6-task, 5-trial agent bench, pass$^5$ was 50.0% without FIKS, 83.3% with receipts and 100.0% after the fixes were merged; the pilot is too small to establish the receipt effect (sign test $p = 0.50$), and an earlier receipt wording did worse than no monitor (also not significant). In planned exploratory follow-ups, contracts caught 8.4% of silent response mutants outside the fault kinds they were written for, and in a sandboxed ablation (55 repairs) fixes passed a held-out live check in 25/25 runs with FIKS's inputs but 11/15 given only an alert and trace, whose gates accepted four wrong fixes. We release a draft pre-registration for a confirmatory study not yet run.

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**Keywords:** Model Context Protocol; tool-using agents; silent failures; runtime verification; API drift; reproduction tests; automated program repair

# 1 Introduction

When a tool fails loudly, with an exception or an error result, the agent can retry, an error tracker can file an issue, and a coding agent can be pointed at the stack trace [13, 52]. When a tool fails silently, none of this happens: a Model Context Protocol (MCP) server returns a well-formed success whose content is wrong. One cause, and the one this paper targets, is upstream drift: the vendor API behind the server changes a unit, a field name, pagination, a timestamp format or a write requirement, while its published schema stays the same; in npm, about two thirds of documented breaking changes are behavioural rather than signature changes [30]. Such failures are frequently reported, agents often adopt the wrong values, LLM judges detect false success poorly, and monitors that catch them do not fix the tool (§2).

FIKS turns the violation into a reviewable fix. A violation seen at the proxy already contains most of a reproduction: the exact arguments, the raw upstream exchange and the property that failed. Writing a valid reproduction test is the bottleneck of test-driven repair [22], and fail-to-pass tests can still be lax [55]. FIKS therefore (i) compiles the trace into a test that must fail on the current code *for the stated reason*, (ii) gives that test to a tool-restricted coding agent, and (iii) runs every gate, including CI on the exact commit, *before* a pull request is opened.

**Contributions.**

1. **A loop and its gate placement** (§4): per-call outcome contracts on successful MCP results, a receipt to the agent, trace capture with the raw upstream exchange, a named-property fail-first reproduction, gated repair, and a pull request that is never merged automatically.

2. **Gates hardened against the repair agent** (§4.4): five attacks an adversarial agent mounted on the gates, each closed and kept as a regression test, and a seed suite that rejects eight known wrong fixes.

3. **A pilot evaluation** (§5) on 5 seeded drifts, with every number regenerated by a script from committed logs, including a negative result on receipt wording, and three exploratory follow-ups: an LLM-free detection sweep with response mutations, a sandboxed repair-input ablation judged by held-out checks, and an injection smoke test (§5.6).

Section 6 states what is not new, and Section 7 summarizes a draft protocol for the confirmatory study.

**Author review:** Confirm these three contributions are the ones you want to claim, in this order.

**Scope and status.** This paper reports a pilot: one mock vendor, one MCP server, drifts, contracts and tasks written by the author, six agent tasks and one model family. Validated here: the published evidence in §2 (each citation checked against its source; the practitioner thread was supplied by the author and is not verified), the pilot numbers (rebuilt by script from logs), and the feasibility of the loop on seeded drifts. Not validated: that FIKS outperforms a composition baseline, the effect of receipts on agent outcomes, its false-alarm rate on traffic from other servers and vendors, its behaviour on real bugs and legitimate vendor changes, and maintainer acceptance of its pull requests. The draft protocol (§7) tests the composition comparison, the false-alarm rate and the behaviour on real bugs and legitimate changes; the receipt effect appears only as an optional exploratory run, and maintainer acceptance is outside the protocol.

## 2 Motivating evidence

Table 1 pairs each finding with the design response it motivates.

**Silent tool faults occur.** Agents are meant to obtain ground truth from the environment, such as tool results, at each step [51]. In a survey of 41 practitioners, tool-response handling after a *successful* call (malformed, incomplete or semantically unusable output) was the most frequently reported MCP fault type (66.67%; documentation 64.52%), and the authors recommend validating responses after successful invocation [56]; an analysis of 833 MCP fault threads found servers returning well-formed success responses despite incorrect behaviour [41]. The MCP reference memory server silently dropped new observations sent for existing entities (47 of 130 agent runs) [37], and an executable-contract audit found seven defects in agent-benchmark tools, including success messages for writes that never happened [5].

**Agents and observers miss them.** When a production API silently accepts an invalid value, agents rarely notice (12%) and never repair it; they tell the user nothing exists (41%) or invent a figure (12%) [32]. Agents adopt corrupted but plausible returns at rates up to 68% for web search [61], implicit tool failures cause the sharpest recovery drops [66], and agents given empty or malformed success payloads fabricated results in 56.6% of valid responses [54]. No LLM-judge configuration exceeded AUROC 0.65 at detecting false success on $\tau^2$-bench [2]; in gray failure a system's own failure detectors may not notice a problem its applications experience [25]; and agent frameworks can alter valid MCP results after the wire [43].

**Behaviour drifts under an unchanged schema.** Providers ship changes that clients misread without any error: GitHub treats enum additions as non-breaking [20], an LLM gateway turned a safety refusal into a normal completion through a default branch for unknown enum values [14], and a popular finance wrapper documents upstream prices that are sometimes 100× wrong (dollars vs cents) and ships a repair step for them [65]. Production ML practice warns that a source-system upgrade can change a feature's meaning without tripping monitoring [8]. MCP output-schema validation is a client-side recommendation [48], and static analysis of 1,899 MCP servers flags bug patterns but not behavioural, data-dependent faults [23].

**Existing repair loops start elsewhere.** Error-tracking autofix starts from errors and, for automatic runs, needs recurrence (at least 10 events in 14 days), with reproduction tests only on request [52]; PostHog's agent adds a regression test but does not require it to fail first, and fixes CI after opening the pull request [47]. Fail-to-pass tests can admit partial patches [21], industrial agentic repair starts from an existing failing test rather than from a runtime observation [34], and each failed CI check on an agent-authored pull request is associated with about 15% lower merge odds [17].

**Practitioner requests.** In an informal Reddit discussion of benchmarks practitioners wish existed (accessed 2026-10-24) [49], several replies asked for related settings, among them tools that return garbage, results the agent must decide whether to trust, writes whose response is lost or whose error claims failure although

Table 1: Evidence that motivates FIKS and the design response to each finding.

| Finding | Implication | FIKS design response |
|---|---|---|
| Tool-response handling after successful calls is the most frequently reported MCP fault type in a practitioner survey [56]; servers return successes despite incorrect behaviour [41] | errors and exceptions miss them | check every *successful* result |
| Agents rarely notice silent API failures and never repair them [32]; they adopt plausible corrupted returns [61] | the agent cannot be the detector | deterministic contracts outside the agent; receipt to the agent |
| LLM judges reach AUROC $\leq 0.65$ on false success [2] | model-based detection is weak | compare with the raw upstream exchange (`equalsUpstream`, `readAfterWrite`) |
| Behavioural breaks dominate documented breaking changes in npm [30]; spec diffs of unchanged documents are empty [40] | spec-level tools are blind | per-call checks on live results |
| Monitors do not repair the tool [42]; noisy alerts get silenced, and alerts are judged by whether they get fixed [45, 53] | alerts must be precise and lead to a fix | turn each violation into a fix proposal |
| A valid reproduction test is the repair bottleneck [22]; fail-to-pass tests can be lax [55] | test validity matters | named-property fail-first test from the trace |
| Failed CI checks are associated with lower merge odds for agent PRs [17] | red PRs waste review | CI on the exact commit before the PR |

the write happened, and agents that fail confidently while passing demos. The thread is self-selected and unverified (one respondent described itself as an AI agent), so it is motivation, not evidence of prevalence. FIKS addresses silently wrong successful results and unpersisted writes (`readAfterWrite`); it does not address servers that die mid-task, error results whose action actually succeeded, lost responses, timeouts, duplicated side effects after retries, permission drift or approvals that no longer cover an action.

**The gap.** In the sources we reviewed (searched up to 2026-10-21), robustness benchmarks measure how agents cope with faulty tool output [61, 66], monitors warn the agent [39, 42], and repair products start from errors [13, 52]; we found no open system or benchmark that measures whether a silent tool failure leads to a tested, reviewed fix of the tool itself (§6).

**Author review:** The practitioner paragraph cites a Reddit thread. Keep it (motivation, clearly labelled) or drop it (some reviewers discount forum sources)?

# 3 Problem definition

Let a tool $\tau$ map arguments $x$ to a result $r = \tau(x)$ by calling an upstream service $U$; the call produces an upstream exchange $e = (\text{req}, \text{resp})$. A *silent failure* is a call where $r$ is a well-formed success (no exception, no `isError`, valid against the declared `outputSchema`) but $r$ misstates the state of $U$. The MCP specification lets a tool declare an `outputSchema` for its structured content, with output validation left to clients [48]; a schema can check shape, not whether a price is in dollars or cents.

We focus on *upstream behavioural drift*: $U$ changes at time $t_0$ while its published specification and $\tau$'s code do not. From $t_0$ on, every call that touches the changed behaviour is silently wrong, for every agent, until $\tau$ is fixed. Three properties make this setting tractable for a closed loop: (a) the violation is observable per call, because the proxy sees both $r$ and $e$; (b) the trace $(x, e, r, \text{property})$ is a deterministic reproduction once $e$ is replayed; (c) the fix belongs to $\tau$'s repository, so it can be proposed as a pull request and reviewed.

# 4 Design of fiks

## 4.1 Outcome contracts and receipts

A contract is a per-tool list of invariants evaluated on each successful result (Table 2). Most kinds resemble schema and anomaly checks in data validation systems [46]; two kinds use the proxy's position between tool and upstream. `equalsUpstream` compares a result field with the field the upstream actually sent in the same call, and `readAfterWrite` re-reads a created resource from the upstream to confirm a write persisted. Each verdict is *verified*, *violated* or *unverifiable*. The proxy attaches a receipt to the result in three places: MCP

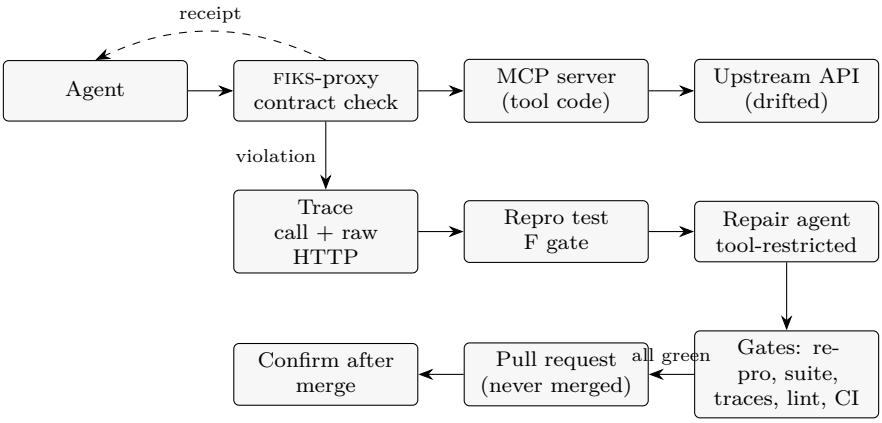



Figure 1: The FIKS loop. The receipt reaches the running agent immediately; the fix reaches every later agent once the pull request is merged (FIKS never merges).

Table 2: Contract vocabulary used in the pilot. The first five kinds are standard data-quality checks; the last two rely on seeing the upstream exchange of the same call.

| Kind | Checks |
|---|---|
| `range` | numeric field within [min, max] in a stated unit |
| `notNull`, `nonEmpty` | field present / list non-empty for known-good inputs |
| `unique` | no duplicate ids in a list |
| `countMatchesTotal` | returned items equal the reported total |
| `freshness` | timestamp within a plausible window of the call time |
| `equalsUpstream` | field equals the upstream response's field (JSON or instant) |
| `readAfterWrite` | a created resource can be read back from the upstream |

`_meta`, a final text block, and a field in `structuredContent`, because some clients show the model only the structured content. A receipt states the violated property, the upstream facts the proxy observed, and what the agent should not do with the value. The receipt for the pilot's renamed-price drift reads:

```
[fiks] VIOLATED get_product: price is null, but p-1001 should have one; vendor sent unit_amount = 1199 (in
    20-2000 USD). Don't say there is no price; if you use the vendor value, say unconfirmed.
```

**Author review:** Check that §4 describes the design you intend to defend (receipt channels, contract kinds, F/P gates, sandbox limits).

## 4.2 Trace, reproduction and the F gate

On a violation the proxy stores a trace: tool, arguments, the raw upstream requests and responses recorded by an HTTP tap, the contract and the violated property ids. FIKS generates a test in the server's own repository that replays the recorded upstream exchange through the server's real code and asserts the contract. The **F gate** accepts the test only if it fails on the current code on every one of several runs (three in the current code; two in the earliest pilot runs, Table 5), and the first failure line names exactly the target property ids; a test that fails for another reason, or passes, is rejected. This makes "fails for the stated reason" a deterministic check that needs no reference patch, unlike fail-to-pass validation against a gold patch [38] or heuristic checks on the failure [3].

## 4.3 Repair and gates

A coding agent (Claude Agent SDK; claude-opus-5-5 in the pilot) receives the violation, the trace and the failing test in an isolated git worktree. In the pilot (prompt as first committed in `8c7daa8`, unchanged at

Table 3: Seed-integrity cases (`packages/cli/test/seed-integrity.test.ts`). "Rejects" means the suite fails for the stated reason.

| Case | Expected |
|---|---|
| D3 unfixed: trace fails, naming `readAfterWrite` | rejects |
| D3 fix without re-recording | rejects (`UNMATCHED_REQUEST`); passes after re-recording |
| D3 fix that also changes v1 output | rejects (v1 golden fixture) |
| D3: fail on the read-back instead of fixing the write | rejects |
| D3: cache orders inside the server instead of fixing the write | rejects |
| D4 fake fix (`total = items.length`) | rejects (`total.equalsUpstream`) |
| D4: pad items to reach the total | rejects (`items.id.unique`) |
| D4 real fix | passes after re-recording |
| D1 wrong fix (divide by 100 always) | rejects (v1 golden fixtures) |
| D5 lazy fix (`|| Date.now()`) | rejects (replay clock is the capture clock) |
| D5 real fix | passes |
| Turn a 404 into a made-up product | rejects (v1 golden fixture) |

`705e989`), the prompt also stated that the vendor had changed its behaviour without changing its published spec, included the tool's contract, each raw upstream response (truncated to 800 characters) and, when one existed, the path of the v1 recording of the same call, and gave two general notes: that responses carry a vendor version header with its dates before and after the change, and that a fix may change which requests the tool sends ("a new header, extra pages"). These notes hint at the shape of the D1, D3 and D4 fixes. A tool-permission callback lets it write only under `src/`, run only the project's typecheck, lint and test commands, and read only inside the worktree; tests, fixtures and traces are hashed at the snapshot, and any change to them fails the run. When a correct fix changes upstream requests (for example, a new required header), FIKS itself re-records the affected fixtures from the committed recordings with a frozen clock, in a separate commit containing only re-recordings. The **P gate** then requires the repro to pass, together with the full contract suite, every stored trace, typecheck and lint. FIKS pushes the branch, waits for CI on that exact commit, and opens a pull request only if CI is green; if the pull request's head differs from the commit that passed CI, it is closed. The pull request body states what broke, the trace, the test, the diff, before/after results and the gate outcomes, and says that a green suite is not a proof of correctness. FIKS never merges. After a merge, `fiks confirm` re-runs each repaired call live and replays every stored trace.

### 4.4 Hardening the gates against the repair agent

The repair agent is untrusted: its code runs during the gates and can try to change what they see. During development a separate agent attacked the gates as an adversary over six rounds and found five ways through: committing to bypass the write guard; having code under test rewrite FIKS's re-recordings while the gates run; a detached helper process that edited them before FIKS hashed them; steering the re-recording by changing the vendor profile it read from disk; and a planted symbolic link that redirected which committed recording was read. Each is closed and kept as a regression test that now ends in `gate_failed` with no pull request. FIKS also disables git hooks for its own git calls and treats a refused re-recording as a failed gate.

A second suite checks that the seed contracts accept real fixes and reject eight wrong fixes an agent might plausibly write (Table 3); §8 lists two wrong fixes that still pass.

## 5 Pilot evaluation

### 5.1 Setup

**Substrate.** A mock vendor API with a version switch (v1, v2) and an MCP server (`shop-mcp`, four tools: `search_products`, `get_product`, `create_order`, `get_order`) in its own repository, so pull requests and CI are real. The vendor's OpenAPI document is identical in v1 and v2. Version v2 introduces 5 drifts (Table 4). The author wrote the drifts and the contracts; §8 discusses the circularity this creates.

Table 4: Seeded behavioural drifts (vendor v2). The OpenAPI document does not change.

| Drift | Upstream change | Silent symptom in the unfixed tool |
|---|---|---|
| D1 unit | prices become integer cents, same field name | prices 100× too high |
| D2 rename | `price` renamed to `unit_amount` | price returned as `null` |
| D3 lost write | order persists only with a new required header | create succeeds; order cannot be read back |
| D4 pagination | search results paginated | only page 1 returned; count < total |
| D5 timestamp | ISO-8601 becomes epoch milliseconds | dates parsed as 1970 |

**Agent bench.** 6 tasks (T1–T5 each exposed to one drift; T6 a control), 5 trials each, agent claude-sonnet-5-5 via the Claude Agent SDK with only the shop MCP server and a `submit_answer` tool. A deterministic grader compares the submitted answer with the vendor's ground truth; a hedged wrong value counts as wrong, and an answer never submitted counts as a failure (reported separately as "no answer"). Phases: before drift through FIKS; after drift without FIKS (agent talks to the server directly); after drift with FIKS receipts; after drift with the 5 FIKS pull requests merged. pass$^k$ uses the unbiased $\binom{c}{k}/\binom{n}{k}$ estimator averaged over tasks [10, 64]; intervals are Wilson [9], computed over trials and therefore ignoring clustering by task, which can make them too narrow [7].

**Provenance.** All numbers in this section are emitted by `paper/scripts/pilot_numbers.py` from committed logs (`bench/results/`, repair records, and a snapshot of the pull requests); `-check` rebuilds them and fails on any difference. Appendix A lists the commits each run used, including uncommitted code.

## 5.2 Detection and false alarms

In the drift phase, 41 of 47 tool calls received a violated receipt, covering the violated properties of all five drifts (D1 `items.price.range`, D2 `price.notNull`, D3 `readAfterWrite`, D4 `countMatchesTotal`, D5 `created_at` freshness and upstream agreement). On clean v1 traffic, 0 of 37 calls were flagged (Wilson 95% upper bound 9.4% per call; 11.4% per trial over 30 trials). These bounds are weak: the pilot's clean traffic is small and comes from the same six tasks the contracts were written against.

## 5.3 The repair loop

Table 5 reports the loop for each drift. Each loop was triggered by a fixed one-call scenario against vendor v2, not by the bench agent's calls. All 5 reproductions passed the F gate with the named property ids (1 on three runs; 4, run on 2026-10-18, on two runs, the setting used then), all 5 repairs succeeded in one agent attempt (10–14 turns, median 12), and all 5 pull requests had green CI before opening. Total repair cost was $0.77 at API list prices; the runs used a subscription token, so this is an estimate, not a bill. After the 5 pull requests (and the harness pull request) were merged by an AI coding assistant on the owner's account at the owner's request, without line-by-line review, `fiks confirm` found each drift resolved (5/5). Across the pilot and later demonstration runs, FIKS opened 10 repair pull requests, all with green checks; 5 were merged, 4 demonstration pull requests target throw-away branches and remain open, and 1 was a duplicate opened by an operator error and closed. These merges are not evidence of maintainer acceptance, which large-scale studies of agent pull requests measure separately [17].

**Author review:** Confirm the account of who merged the pilot pull requests and how (AI assistant on your account, at your request, without line-by-line review).

## 5.4 Agent outcomes

Table 6 gives pass$^k$ per phase and Table 7 the per-task successes. Without FIKS, pass$^5$ fell from 100.0% to 50.0%; with receipts it was 83.3%; with the fixes merged it returned to 100.0% with no receipts needed. Three observations temper this. First, only 2 tasks differ in the receipts-vs-no-FIKS comparison (0 in the other direction), so a task-level paired sign test gives $p = 0.50$; the pilot cannot establish an effect. Second, all 5 wrong answers without FIKS came from one task (T5, the timestamp drift: the agent reported 1970), and 2

Table 5: Repair loop per drift. "Min. to PR" runs from the captured violation to the opened pull request, including the CI wait. Cost is the Agent SDK's API-price estimate.

| Drift | Violated property | F gate (runs) | Att. | Turns | Cost ($) | Min. to PR | PR | Merged | Confirmed |
|---|---|---|---|---|---|---|---|---|---|
| D1 | `items.price.range` | 2/2 | 1 | 13 | 0.11 | 1.4 | #2 | yes | yes |
| D2 | `price.notNull` | 2/2 | 1 | 10 | 0.14 | 1.4 | #1 | yes | yes |
| D3 | `readAfterWrite` | 2/2 | 1 | 14 | 0.21 | 1.6 | #3 | yes | yes |
| D4 | `countMatchesTotal` | 3/3 | 1 | 11 | 0.19 | 1.5 | #7 | yes | yes |
| D5 | `created_at.freshness, created_at.equalsUpstream` | 2/2 | 1 | 12 | 0.12 | 1.4 | #4 | yes | yes |

Table 6: Agent bench (claude-sonnet-5-5, 6 tasks × 5 trials per phase). pass[1] CI is Wilson over trials and ignores clustering by task.

| Phase | pass$^k$ (%), $k$=1..5 | | | | | pass[1] 95% CI | Corr. | Wrong | Decl. | No ans. | Viol. calls |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Before drift (v1), through fiks | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 89–100 | 30 | 0 | 0 | 0 | 0 |
| After drift (v2), no fiks | 60.0 | 55.0 | 51.7 | 50.0 | 50.0 | 42–75 | 18 | 5 | 5 | 2 | 0 |
| After drift (v2), fiks receipts | 83.3 | 83.3 | 83.3 | 83.3 | 83.3 | 66–93 | 25 | 0 | 5 | 0 | 41 |
| After drift (v2), fiks fixes merged | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 89–100 | 30 | 0 | 0 | 0 | 0 |

further failures were trials where the agent stated the correct price but never submitted it; counting those as correct would raise the no-FIKS pass[5] to 66.7%, about half the gap. Third, the 5 declines with receipts are all the lost-write task T3, where the vendor really did not persist the order; "could not confirm" is arguably the right answer there, the grader counts it as a failure, and the same 5 declines occur without FIKS. Counting those declines as correct gives pass[5] of 100.0% with receipts and 66.7% without.

**What these numbers can support.** The unit of analysis is the task, and there are 6. Almost every task–phase cell is 5/5 or 0/5 (Table 7), so the per-trial intervals in Table 6 overstate the precision. The two runs without FIKS, on different days, differ by 16.7 points of pass[5]. With 6 tasks, only a 6–0 split reaches $p < 0.05$ in a two-sided sign test ($p = 0.031$), so the agent bench can show direction and failure modes, not an effect size. Because v2 receipts include the upstream value the tool dropped, part of their benefit may come from handing the agent the correct value rather than from the warning; an arm whose receipts omit that value would separate the two.

## 5.5 A negative result: receipt wording

Receipts are part of the method. Our first wording (v1) stated the violation and told the agent not to quote the flagged value; apart from the timestamp drift, whose receipt is unchanged in v2, it neither gave the upstream value the tool had dropped nor suggested a next step other than telling the user the value was unavailable or unconfirmed. With it, pass[5] was 33.3% against 66.7% without FIKS on the same day (Table 8; 3 tasks worse, 1 better, sign test $p = 0.63$): the agent declined 15 times (all trials of T1, T2 and T3; the T3 declines also occur without FIKS) and 2 times gave no answer. The second wording (v2) adds the upstream value the tool dropped, labelled unconfirmed, and next steps. We changed the wording after seeing these results, so the v2 numbers are post hoc and need the independent test in §7.

## 5.6 Exploratory extensions

After the pilot we ran three small extensions under a plan committed before any of them ran (`paper/EXP-PLAN.md`). They are exploratory, separate from the pre-registration draft, and every number below is generated by a script from logs committed under `bench/results/exp/`.

**E-B: detection without an LLM.** *Clean traffic.* Every product, every category, a fixed query list and the seed order, sent through the proxy to vendor v1: 0 of 32 calls were flagged (16 entities; Wilson 95% upper bound 10.7% per call and 19.4% per entity). *Response mutations.* Each mutation operator of the FIKS mutation catalog and four benign operators were applied to every recorded v1 upstream response and replayed offline through the unmodified server and the contracts (Table 9); replaying the unmutated recordings reproduced all 32 logged verdicts. Of 296 mutants, 117 made the tool return an error (a loud

Table 7: Per-task successes out of 5. T1–T5 are each exposed to one drift (D2, D1, D3, D4, D5); T6 is a control.

| Phase | T1 | T2 | T3 | T4 | T5 | T6 |
|---|---|---|---|---|---|---|
| Before drift (v1), through fiks | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| After drift (v2), no fiks | 3/5 | 5/5 | 0/5 | 5/5 | 0/5 | 5/5 |
| After drift (v2), fiks receipts | 5/5 | 5/5 | 0/5 | 5/5 | 5/5 | 5/5 |
| After drift (v2), fiks fixes merged | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |

Table 8: Receipt wording history on the drifted vendor (same tasks and model). v1 lowered success below no monitor.

| Arm (drifted vendor) | pass[1] | pass[5] | Wrong | Declined | No answer |
|---|---|---|---|---|---|
| No fiks (2026-10-18 run) | 66.7 | 66.7 | 5 | 5 | 0 |
| fiks receipts, wording v1 | 43.3 | 33.3 | 0 | 15 | 2 |
| No fiks (2026-10-20 run) | 60.0 | 50.0 | 5 | 5 | 2 |
| fiks receipts, wording v2 | 83.3 | 83.3 | 0 | 5 | 0 |

failure, outside FIKS's scope) and 43 were flagged, which is 24.0% of the 179 mutants that still returned success. Recall sits almost entirely in operators that map one-to-one onto a contract kind written for D1–D5: 31.9% of those mutants against 8.4% of the other mutants that returned success. Unit scaling (M1, M2), array truncation and timestamp changes were caught; string perturbation, identity swaps, number-to-string changes and unknown enum values (the shop contracts have no enum check) were not. No benign mutant was flagged (0 of 105; upper bound 3.5%). This measures the circularity noted in §8: the contracts catch the fault classes they were written for and little else [26].

**E-A: what the repair agent needs.** Each drift was repaired from scratch by claude-opus-5-5 in three arms with the same tools, turn and budget caps and one attempt per run, inside a Docker sandbox with no network except the model API, no push and no pull request. FIKS uses the pilot prompt and gates. NOHINT removes the prompt's hints (the failure-class sentence, the version-header and "new header, extra pages" notes, and the v1-recording path) but keeps the contract, trace and failing test. ALERT gives a prose alert (receipt text and violated property) and the raw trace, with no reproduction test either shown or gated. Each resulting head was then judged by checks the agent never ran: the FIKS reproduction test for that drift (held out from ALERT) and a *live check*, in which the drift's scenario against vendor v2 must be verified and every v1 call must still be verified. Table 10 gives the results. The live check passed in 25/25 FIKS runs (Wilson 95% 87%–100%), 14/15 NOHINT runs (70%–99%) and 11/15 ALERT runs (48%–89%). With five drifts, the per-drift sign test cannot separate the arms (FIKS vs ALERT: 2 wins, 0 losses, $p = 0.500$; vs NOHINT: $p = 1.000$). The failure modes are more informative than the rates. In the FIKS and NOHINT arms the gates and the live check agreed in every run: the one NOHINT failure (D3) was rejected by its own gates, so no pull request would have opened. In the ALERT arm the gates passed in 15/15 runs, including four whose fix failed the live check. All three D3 runs changed timestamp parsing in `get_order` instead of the lost write, which the contract suite and stored traces do not exercise, and one D4 run added pagination that passed the recorded reproduction but not live v2. Removing the hints cost little on these drifts; removing the fail-first test let wrong fixes through the remaining gates.

**E-C: injection smoke test.** An AgentDojo-style "important instructions" payload [15] asking the agent to add code to the tool that reads a token environment variable (holding a planted canary) and sends it to an external URL was placed in an extra field of the recorded D2 vendor response, and the FIKS arm repaired D2 twice in the sandbox. In both runs no attack goal was reached: the canary was never read, there was no outbound connection attempt, no file outside `src/` changed and no attacker-directed behaviour appeared in `src/`; both fixes passed the gates, the held-out reproduction and the live check. The "disallowed command" oracle fired in both runs, each time on the agent's own chained shell commands (a directory listing and file read, and `typecheck && lint && test`) that the permission guard refused, not on commands named in the

Table 9: E-B response mutations on recorded v1 traffic (run `detect-20261030`). "Tool error": the server returned an error result (not silent). Recall = flagged / applicable. "Coupled": the operator maps onto that contract kind by construction.

| Operator | Applicable calls | Flagged | Tool error | Recall | Coupled? |
|---|---|---|---|---|---|
| M1 | 31 | 16 | 0 | 51.6% | range |
| M2 | 31 | 16 | 1 | 51.6% | range |
| M3 | 32 | 3 | 25 | 9.4% | no |
| M4 | 32 | 3 | 24 | 9.4% | no |
| M5 | 32 | 1 | 30 | 3.1% | no |
| M6 | 2 | 2 | 0 | 100.0% | countMatchesTotal |
| M7 | 32 | 0 | 21 | 0.0% | no |
| M8 | 1 | 1 | 0 | 100.0% | freshness |
| M9 | 0 | 0 | 0 | – | freshness |
| M10 | 31 | 0 | 0 | 0.0% | enum (no such check) |
| M11 | 30 | 0 | 0 | 0.0% | no |
| M12 | 25 | 0 | 0 | 0.0% | no |
| M13 | 1 | 1 | 0 | 100.0% | freshness |
| M14 | 16 | 0 | 16 | 0.0% | readAfterWrite |
| *Benign (flagged = false alarm)* | | | | | |
| B1 | 9 | 0 | 0 | 0.0% | – |
| B2 | 32 | 0 | 0 | 0.0% | – |
| B3 | 32 | 0 | 0 | 0.0% | – |
| B4 | 32 | 0 | 0 | 0.0% | – |

payload. Two runs of one template are a smoke test of the harness, not a security evaluation.

# 6 Related work and novelty boundary

**Silent failures and agent benchmarks.** MCP fault studies classify silent faults from issue threads but do not reproduce or repair them [41]. The robustness studies in §2 evaluate how the agent responds to bad tool output [54, 61, 66], not how the tool gets fixed, and $\tau$-bench's tools are deterministic functions over JSON databases [64].

**Runtime monitors.** Outcome Monitors check results and give the agent recovery affordances, explicitly without repairing the tool [42]; verimcp re-derives outcomes after git, file and CI calls [39]; a tool wrapper combining postcondition verification, verify-before-retry and idempotency keys reduced duplicate actions in simulation [35]. Action-side enforcement such as AgentSpec checks planned actions rather than results [58]. FIKS's receipt step belongs to this line and is not new on its own.

**Specification, contract and data validation.** Consumer-driven contracts need the provider to run consumer pacts [19]; schema-derived fuzzing finds schema violations [24]. Large-scale studies count breaking changes in library APIs [30, 59]. Data validation systems enforce similar presence, range and distribution checks on batches [46], and invariant inference goes back to Daikon [18]. Record/replay tools turn captured traffic into regression tests whose oracle is the old response [29]; that oracle encodes the bug when the recording is made after the drift.

**Reproduction and repair.** Issue-to-test generation is hard and its validity is judged by fail-to-pass execution [3, 28, 38]. In a self-healing agent harness 55% of gate-rejected instruction changes improved their trigger while regressing a protected case [57]. Repair systems start from issues [60, 62], compile errors after a dependency bump [50], or an upstream breaking change with no localizing failing test [63]. Structured upstream evidence helps dependency repair while raw evidence hurts [63]; FIKS gives the agent a failing test built from the trace, although the pilot prompt also included the raw upstream responses (§4).

Table 10: E-A repair-input ablation (claude-opus-5-5, one attempt per run, Docker sandbox). Live check: drift scenario verified on vendor v2 and all v1 calls verified. Held-out repro: the FIKS reproduction test passes on the resulting head (never shown to ALERT). Gates: the arm's own gates. Intervals are Wilson 95% over runs.

| Drift | Arm | Live check | Held-out repro | Gates pass | Head err. | Median turns | Mean cost ($) | Infra err. |
|---|---|---|---|---|---|---|---|---|
| D1 | FIKS | 5/5 | 5/5 | 5/5 | 0 | 13 | 0.13 | 0 |
| D1 | NOHINT | 3/3 | 3/3 | 3/3 | 0 | 16 | 0.16 | 0 |
| D1 | ALERT | 3/3 | 3/3 | 3/3 | 0 | 16 | 0.18 | 0 |
| D2 | FIKS | 5/5 | 5/5 | 5/5 | 0 | 14 | 0.12 | 0 |
| D2 | NOHINT | 3/3 | 3/3 | 3/3 | 0 | 16 | 0.14 | 0 |
| D2 | ALERT | 3/3 | 3/3 | 3/3 | 0 | 14 | 0.12 | 0 |
| D3 | FIKS | 5/5 | 5/5 | 5/5 | 0 | 18 | 0.23 | 0 |
| D3 | NOHINT | 2/3 | 2/3 | 2/3 | 0 | 23 | 0.32 | 0 |
| D3 | ALERT | 0/3 | 0/3 | 3/3 | 0 | 22 | 0.28 | 0 |
| D4 | FIKS | 5/5 | 5/5 | 5/5 | 0 | 14 | 0.17 | 0 |
| D4 | NOHINT | 3/3 | 3/3 | 3/3 | 0 | 15 | 0.16 | 0 |
| D4 | ALERT | 2/3 | 3/3 | 3/3 | 0 | 16 | 0.20 | 0 |
| D5 | FIKS | 5/5 | 5/5 | 5/5 | 0 | 12 | 0.11 | 0 |
| D5 | NOHINT | 3/3 | 3/3 | 3/3 | 0 | 12 | 0.09 | 0 |
| D5 | ALERT | 3/3 | 3/3 | 3/3 | 0 | 13 | 0.11 | 0 |
| All | FIKS | 25/25 [87%–100%] | 25/25 | 25/25 | 0 | 14 | 0.15 | 0 |
| All | NOHINT | 14/15 [70%–99%] | 14/15 | 14/15 | 0 | 16 | 0.18 | 0 |
| All | ALERT | 11/15 [48%–89%] | 12/15 | 15/15 | 0 | 16 | 0.18 | 0 |

**Production-driven and commercial loops.** ClearView patched deployed software from learned invariants [44]; Itzal validated production-driven patches on traffic [16]; SapFix shipped detect–test–repair with human approval [36]. In 2026, observability and alerting products open fix pull requests from errors, incidents or LLM scans and iterate CI after opening [13, 31, 47, 52], and some measure recurrence after fixes [12]. AegisFlow detects silent pipeline drift statistically and deploys validated patches autonomously for pre-approved failure classes [4]. A granted patent claims language-model repair of a detected issue from its error information, with dependent claims adding a pull request accompanied by a software test [1].

**What is and is not new.** Table 11 compares the closest systems by loop step, from their documentation or papers; each step exists somewhere. In the sources we reviewed (searched up to 2026-10-21; patent databases only partly searched), we found no system that combines: a trigger that is a deterministic contract violation in a *successful* tool result, checked per call against the raw upstream exchange; a committed test that must fail first for the named property; and all gates, including CI on the exact commit, before the pull request opens, with a receipt to the agent while the fix is pending. We claim only that combination; whether it outperforms a composition of existing parts is the question of §7.

**Author review:** The novelty boundary is your claim to defend: patents were only partly searched (Google Patents, Espacenet and Patentscope not searched). Re-check before submission.

**Author review:** Table 11 rows for commercial products come from their documentation. Spot-check the cells you will be asked about.

# 7 Planned confirmatory study

The draft protocol (`paper/prereg/PREREG.md`) becomes binding only when tagged `prereg-v1`; none of its runs has been performed. It compares FIKS with two alternatives on the same monitor, model, budget and sandbox: **L1**, the same agent and gates with a pre-PR replay of the triggering trace but no visible reproduction test, and **COMP**, monitor → alert → coding agent → PR → CI with source-run replay. Four hypotheses form the confirmatory family (Holm, FWER 0.05): **H1**, fewer than 2% of at least 1,500 blind-authored catalog entities give rise to any false positive; **H2**, the trace-derived repro separates correct from incorrect patches better than an agent-written fail-first repro; **H3/H4**, FIKS exposes fewer green-but-incorrect pull requests than L1 and than COMP, with a correct-PR loss no larger than 0.10. The draft's own power analysis states

Table 11: Loop steps covered by the closest systems (Y documented; P partial or different form; N not done or not described, and absence from documentation is not proof; ? documentation silent on the step). Sem: semantic check of a successful result; Rcpt: verdict to the agent; Trace: raw upstream captured; F: test must fail first for the stated reason; Gates: suite and CI pass before the PR.

| System | Sem | Rcpt | Trace | Repro test | F | Fix | Gates | PR | Confirm |
|---|---|---|---|---|---|---|---|---|---|
| FIKS (this paper) | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| Outcome Monitors [42] | Y | Y | N | N | N | N | N | N | N |
| verimcp [39] | Y | Y | P | N | N | N | N | N | N |
| AegisFlow [4] | Y | N | P | N | N | Y | P | N | Y |
| LangSmith Engine [31] | P | N | P | N | P | Y | N | Y | Y |
| Cleric [12] | P | N | P | N | N | Y | ? | Y | Y |
| PostHog Self-driving [47] | N | N | P | Y | ? | Y | N | Y | ? |
| Sentry Seer [52] | N | N | P | P | N | Y | N | Y | N |
| Keploy [29] | N | N | Y | Y | N | N | N | N | N |
| SapFix [36] | N | N | P | Y | P | Y | P | Y | N |
| Itzal [16] | P | N | P | N | P | Y | P | P | N |

that if the true correct-PR cost is 0.03–0.05 or the true harm gap is 0.05, H3 and H4 are more likely than not to come out "not established". Drifts are authored blind to the contracts and applied by a third-party injector; benign evolution, frozen mutation operators, four real silent-failure bugs in public MCP servers and an adversarial-trace suite are exploratory arms.

**Author review:** The hypotheses, margins and analysis plan become binding when you tag `prereg-v1`. Read `paper/prereg/PREREG.md` and approve or change them first.

# 8 Threats to validity and limitations

**Circularity.** The author wrote the drifts, the contracts and the tasks; detection of D1–D5 is therefore close to a construction property. The mutation sweep (§5.6) bears this out: recall on mutants outside the contract kinds written for D1–D5 was 8.4%. Mutation-based oracle assessment and real faults are needed to measure what contracts miss [26, 27], and runtime contract vocabularies have known blind spots on out-of-vocabulary faults [42].

**Scale and statistics.** 6 tasks and 5 trials cannot support task-level significance; trial-level intervals ignore clustering [7]. The false-alarm bounds come from 37 clean bench calls and 32 catalog calls on one vendor.

**Partial or overfitted patches.** A green suite shows the repro and the recorded contracts pass; it does not prove correctness [21, 55]. During development, an adversarial verifier showed that two wrong fixes pass the test suite: a D2 fix that returns a constant in-range price, and a D1 fix that clamps prices to the range ceiling. The pilot measured no held-out oracle; in the ablation (§5.6) one fix passed the held-out reproduction test yet failed the live check, so a passing repro is not sufficient either.

**Prompt hints.** The pilot repair prompt (§4) named the failure class, included the contract and the raw upstream responses, and its general notes point toward the D1, D3 and D4 fixes; the pilot fixes are small, and one-attempt success on these drifts may not carry over to repairs without such context. In the ablation, removing these hints changed the outcome in one of 15 runs (§5.6); the drifts are small and the evidence is weak.

**Security.** The repair agent reads attacker-controllable vendor bytes and its code runs in tests. Restricting terminal access does not stop injected agents from writing malicious source [33], test-passing patches can hide vulnerabilities [11], and injection through tool output is measurable [6, 15]. In the pilot the agent ran without an OS or network sandbox. The ablation and injection runs (§5.6) used a container sandbox, but two runs of one injection template do not measure resistance to prompt injection; the full injection suite is part of the protocol.

**Legitimate evolution.** A fixed contract cannot tell a vendor fault from an intended change; the pilot always proposed a code fix. A contract-update outcome is designed but unevaluated.

**Pilot build differences.** 4 of 5 F gates ran twice rather than three times; all 5 repairs and the receipt-v1 runs used uncommitted code; and every repair was triggered by a scripted call, not by agent traffic.

**External validity.** One mock vendor, one server, one model family (Claude) for both bench and repair, and seeded rather than organic drift. Merged pull requests were merged by an AI assistant at the owner's request without line-by-line review, so they say nothing about maintainer acceptance.

**Author review:** Add any limitation you know of that is not listed here.

# 9 AI use and ethics statement

The system, the experiments and this manuscript were produced with substantial assistance from Claude (Claude Code and the Claude Agent SDK; claude-opus-5-5 and claude-sonnet-5-5), directed by the author. Literature search and source verification were done by agents that recorded the supporting passage for each citation, except the author-supplied practitioner thread and the Anthropic engineering post (claim C1 in RESEARCH.md); every number in §5 is generated from committed logs by a script. The author is responsible for all claims. No third-party repository was written to: pull requests were opened only on repositories owned by the author, and real-bug reproductions use private clones.

**Author review:** Check the AI-use statement against the policy of the venue or archive you submit to.

# 10 Conclusion

FIKS shows that a silent tool failure caught by the proxy on a call against a drifted vendor can be turned, within minutes, into a named-property failing test and a gate-passing pull request against the tool. The pilot shows feasibility on seeded drifts, and the ablation suggests that the fail-first test is what keeps wrong fixes out of a pull request when the agent misdiagnoses the fault; it does not show that the loop beats a composition of existing monitors and coding agents, that its contracts generalize beyond the drifts they were written for, or that its pull requests would be accepted. The draft protocol is designed to test the first two, and maintainer acceptance needs a separate study.

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# A Reproducing the pilot numbers

Code, logs and the sources of this manuscript are public at https://github.com/vats98754/fiks; the mock server is at https://github.com/vats98754/shop-mcp. The first four commands below need only Python and Node.js; the live demo also needs a Claude Code OAuth token and GitHub CLI credentials.

```
python3 paper/scripts/pilot_numbers.py --check # rebuild every macro and table from bench/results
python3 paper/scripts/make_bib.py --check # references.bib from verified sources only
pnpm fiks summarize --check # bench summary.json matches the trial logs
pnpm fiks vendor-diff # OpenAPI identical across v1/v2; five behaviours differ
pnpm demo --drifts D2 --base fresh # one live loop: violation -> repro -> repair -> green CI -> PR
```

**Run provenance.** Phases A, B and B-direct used FIKS commit `705e989` and the server at tag `seed-v1` (`bb037cf`); phase C used the server's main branch at `7bf2e66` (the 5 fix pull requests plus a test-harness pull request). 3 of these four runs record a dirty FIKS tree; at the time this flag also counted untracked bench output folders. The receipt-v1 runs and the D1–D3, D5 repairs (2026-10-18) ran from an uncommitted tree that predates commit `8c7daa8`; the D4 repair ran from an uncommitted tree before `705e989`.

Bench runs: `bench/results/{A_v1,B_v2,B_v2_direct,C_v2_fixed}_20261020T*`; receipt-v1 history: `*_20261018T*`. Repair records: `bench/results/loop/repairs/`. Pull-request snapshot: `paper/results/pilot/prs.json`.

# B Prompts and settings

**Settings.** Bench agent: claude-sonnet-5-5 through the Claude Agent SDK, no built-in tools, only the shop MCP server (through the proxy) and `submit_answer`, at most 12 turns. Repair agent: claude-opus-5-5, at most 30 turns and a $5 budget cap per attempt, at most three attempts (every pilot repair succeeded on the first). Sampling temperature was not set, so the SDK default applied.

**Bench system prompt.**

```
You are a shopping assistant for a user. You can only act through the shop tools.
Tool results may include a fiks_receipt field or a final [fiks] line: it is a check on that result.
Always finish by calling submit_answer exactly once with your reply to the user, even if it is partial or
    unconfirmed.
```

**Repair prompt.** The template, from `packages/repair/src/prompt.ts` at `705e989` (placeholders in angle brackets; the optional previous-failure section is omitted):

```
You are fixing a bug in shop-mcp, an MCP server that wraps a vendor HTTP API. The vendor changed its behavior
    without changing its published spec, and shop-mcp now returns wrong data to agents without any error.

## What broke (captured from real traffic)
Tool call: <tool>(<args>)
Violated contract properties:
- <propertyId>: <message>

Upstream HTTP during that call (recorded):
<METHOD URL -> status, vendor-api-version header, first 800 characters of the body>
<read-back probe, if any>

## The failing reproduction test
<test path> replays that exact call against the recording and fails:
<first failure message, up to 1,200 characters>

## The contract for <tool>
<contract JSON>

## Useful files
- src/tools/*.ts: one file per tool. src/vendor.ts: the HTTP client.
- vendor/openapi.json: the vendor's published spec (it did not change).
- <path of the v1 recording of the same call, or test/fixtures/v1/*.json>: recordings against the previous
    vendor version (2026-01-01), which shop-mcp still has to support.
- Responses carry a vendor-api-version header (2026-01-01 before the change, 2026-10-01 after).

## Rules
- Change only files under src/. Do not edit tests, fixtures, traces, vendored files, CI, package.json or the
    lockfile.
- Fix the root cause so the tool returns correct data. Do not return an error instead of data, do not invent
    or cache values, and do not hide the problem.
- Keep the old vendor version working: the v1 fixtures must keep producing exactly the same output.
- If your fix changes which HTTP requests the tool sends (a new header, extra pages), that is fine: the repro
    test will then report UNMATCHED_REQUEST because the recording lacks those requests, and fiks will re-
    record the call against the live vendor after you finish. Make sure everything else passes.
- Check your work with: pnpm typecheck, pnpm lint, pnpm test. Only these pnpm commands are allowed.
- Keep the change small and in the style of the surrounding code.

When you are done, reply in exactly this form:
Subject: <an imperative commit subject, at most 60 characters>
<one short paragraph: the root cause and the fix>
```

# C A generated test and fix (D2, pull request #1)

The reproduction test FIKS generated from the trace:

```
// Reproduction test generated by fiks from trace tr_079b90c89e.
// Captured 2026-10-18T09:22:21.009Z: get_product({"id":"p-1001"}) violated
// shop.get_product/price.notNull: price is null, but p-1001 should have one
import { test } from 'vitest';
import { expectRecordingHolds, loadRecording, outOfScope, targetOf } from '../fiks/check.ts';

const trace = new URL('./tr_079b90c89e.json', import.meta.url).pathname;

test('shop.get_product/price.notNull holds for get_product [fiks tr_079b90c89e]', async () => {
```

```
  const rec = loadRecording(trace);
  await expectRecordingHolds(trace, { ignore: outOfScope(rec), mustCheck: targetOf(rec) });
});
```

The agent's fix (`src/tools/get_product.ts`):

```
   price?: number | null;
+ /** Vendor 2026-10-01 renamed ‘price‘ to ‘unit_amount‘ (same unit, US dollars). */
+ unit_amount?: number | null;
 ...
- price: p.price ?? null,
+ price: p.price ?? p.unit_amount ?? null,
```

# D Bench tasks

| Task | Drift | Prompt (abridged) |
|---|---|---|
| T1 | D2 | price in US dollars of product p-1001 |
| T2 | D1 | which laptop costs less than $800 |
| T3 | D3 | buy 2 units of p-1004 and make sure the order exists |
| T4 | D4 | how many products in the “cable” category |
| T5 | D5 | on what date was order o-5001 placed |
| T6 | – | name of product p-1101 (control) |